

Homework assignment for 02YELMA

Chapter: Magnetostatics

1. Find the magnetic field in the center of a square loop, which carries a steady current I . Let R be the distance from center to side (Hint: You may make use of the result we obtained in the lecture about the magnetic field created by a finite wire).
2. A steady current I flows down a long cylindrical wire of radius a . Find the magnetic field, both inside and outside the wire, if the current is uniformly distributed over the outside surface of the wire.
3. A circular wire loop, with radius R , lies in the xy plane, centered at the origin, and carries a current I running counterclockwise as viewed from the positive z axis.
 - (a) What is its magnetic dipole moment?
 - (b) What is the approximate magnetic field at points far from the origin?
 - (c) Recall the exact magnetic field expression derived for a circular loop for the z axis by using the Biot-Savart law. Show that, for points on the z axis, the answer you found for (b) is consistent with the exact field when $z \gg R$.
4. Consider Rutherford's atomic theory: A positive charge $(+e)$ is located in the center, around which a negative charge $(-e)$ orbits in a circular motion with radius R and linear velocity v . Although technically this orbital motion does not constitute a steady current, in practice the period of motion is so short that it can be approximated as a steady current. If the negative charge rotates counterclockwise, as observed from the positive z axis, what is the magnitude and direction of the orbital magnetic dipole moment in terms of e , v , and R ?
5. Consider a uniformly magnetized sphere with magnetization $\vec{M} = M\hat{z}$ and radius R .
 - (a) Calculate volume and surface bound currents, $\vec{J}_b$ and $\vec{K}_b$. (Hint: Consider the spherical coordinates while calculating the cross products)
 - (b) Think of a spherical shell with radius R rotating about z axis with constant angular velocity ω . Note that the linear velocity of a point on the surface $\vec{v}$ must be a function of θ in spherical coordinates (as θ increases, the point on the sphere covers more distance per unit of time). Express $\vec{v}$ in terms of ω , R , θ (don't forget its direction).
 - (c) If the spherical shell in (b) has a surface charge density σ , find its surface current density $\vec{K}$.
 - (d) Compare $\vec{K}_b$, the bound surface current obtained for a uniformly magnetized sphere, and $\vec{K}$, the surface current density of a rotating spherical shell with angular velocity ω . Explain the analogy between these two cases. What is the relationship between M and σ , ω and R ?
 - (e) Write down the magnetic dipole moment $\vec{m}$ of the magnetized sphere.

6. Utilize the insights gained from the preceding question regarding the magnetized sphere and the rotating spherical shell with a surface charge density to determine the equatorial linear velocity of an electron (assuming that electron is a rotating spherical shell). Refer to the classical radius for the electron as found in Homework 4. Compare your result with the speed of light.

Given for 3: The magnetic field created by a pure magnetic dipole:

$$\vec{B}_{dip}(r, \theta) = \frac{\mu_0 m}{4\pi r^3} \left[2 \cos(\theta) \hat{r} + \sin(\theta) \hat{\theta} \right]$$